25900053 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCCFS101 - FUNDAMENTALS OF BLOCK CHAIN AND CRYPTOCURRENCY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **1** marks

1. Discuss the role of partition tolerance in a distributed system. [U] / [CO1]
2. Discuss about decentralized system. [U] / [CO1]
3. Discuss the role of consensus algorithms in blockchain. [U] / [CO1]
4. Explain the concept of decentralization in blockchain technology. [U] / [CO1]
5. Define and examine asymmetric cryptography. [An] / [CO2]
6. Examine the purpose of Bitcoin Improvement Proposals (BIPs) in the blockchain ecosystem. [An] / [CO3]
7. Discuss about block cipher. [U] / [CO1]
8. Distinguish between Proof of Work (PoW) and Proof of Stake (PoS) consensus algorithms. [An] / [CO2]
9. Select a reason why merchants should accept Bitcoin payments. [E] / [CO3]
10. State the advantages of using a hardware wallet. [E] / [CO3]

11. Recommend a method to improve the scalability of the Bitcoin network. [E] / [CO3]

12. Rank Bitcoin payment methods based on transaction speed and cost. [E] / [CO3]

[1x10 = 10]

Part B

Short Essay Type Questions

Answer any **3** questions

Each question carries **5** marks

13. Compare the advantages and disadvantages of centralized and decentralized distributed systems? [U] / [CO1]

14. Explain the role of decentralization in blockchain technology and how it transforms traditional system. [U] / [CO1]

15. Prove why Bitcoin is considered a decentralized currency and how it differs from central bank digital currencies (CBDCs). [E] / [CO3]

16. Analyse Confidentiality, Integrity, and Authentication in cryptographic systems. [An] / [CO2]

[5x3 = 15]

Part C

Essay Type Questions

Answer any **1** questions

Each question carries **10** marks

17. Differentiate the different layers of a decentralized ecosystem and explain their functions. [An] / [CO2]

18. Recommend a secure setup process for installing and using a Bitcoin wallet. [E] / [CO3]

[10x1 = 10]